

Code changes from implicit to CEL

fe-large-displacement

Each section lists what you must *add or change* relative to the previous rung. Geometry, rollers, the rigid indenter and the soil block stay put until CEL. Shared builder: `examples/common/indenter_model.py:99`.

1 Implicit

The Standard baseline. `examples/indenter/implicit`.

Geometrically nonlinear static step

Without `nlgeom` there is no large displacement. Automatic stabilisation is fictitious viscous damping. It is on because the analysis otherwise stops as soon as the soil fails. Check ALLSD against ALLIE.

```

53     return model.StaticStep(
54         name=name, previous=previous, description=description,
55         nlgeom=ON if nlgeom else OFF,
56         timePeriod=time_period,
57         initialInc=0.005, minInc=1.0e-8, maxInc=0.05,
58         maxNumInc=1000,
59         stabilizationMethod=DISSIPATED_ENERGY_FRACTION,
60         stabilizationMagnitude=2.0e-4,
61         adaptiveDampingRatio=0.05,
62         continueDampingFactors=ON,
63     )

```

Displacement control

The indenter is driven by a prescribed displacement on the rigid-body reference point, ramped with a tabular amplitude over a unit time period.

```

199         time_period = 1.0
200         steps.implicit_step(model, step_name, 'Initial', time_period,
201                             nlgeom=True)
202         model.TabularAmplitude(name='Ramp', timeSpan=STEP,
203                                data=((0.0, 0.0), (time_period, 1.0)))

```

```

228         model.DisplacementBC(name='Push', createStepName=step_name,
229                               region=asm.sets['IndRP'],
230                               u1=0.0, u2=0.0, u3=-P.ind_depth,
231                               ur1=0.0, ur2=0.0, ur3=0.0,
232                               amplitude='Ramp', distributionType=UNIFORM)

```

Standard element library

Same C3D8R topology as Explicit. The library flag must be STANDARD.

```

150     library = EXPLICIT if solver == 'explicit' else STANDARD
151     # Hourglass control is left at the element-type default here. The
152     # explicit model can change it through *Section Controls.
153     elem = mesh.ElemType(elemCode=C3D8R, elemLibrary=library)

```

Standard general contact

ContactStd, not the Explicit interaction.

```

108     interaction = model.ContactStd(name='GeneralContact',
109                                   createStepName=step_name)
110     interaction.includedPairs.setValuesInStep(stepName=step_name,
111                                               useAllstar=True)
112     interaction.contactPropertyAssignments.appendInStep(
113         stepName=step_name, assignments=((GLOBAL, SELF, prop_name),))
114     return interaction

```

Standard job

No DOUBLE_PLUS_PACK. Thread parallelisation, not Explicit domains.

```

84     return mdb.Job(
85         name=job_name, model=model_name, description=description,
86         type=ANALYSIS,
87         numCpus=n_cpus, numDomains=n_cpus,
88         multiprocessingMode=DEFAULT,
89         userSubroutine=user_subroutine,
90         memory=memory_percent, memoryUnits=PERCENTAGE,
91         echoPrint=OFF, modelPrint=OFF, historyPrint=OFF, contactPrint=OFF,
92         getMemoryFromAnalysis=True,
93     )

```

Optional geostatic

Self-weight belongs in a *Geostatic step, then import into later steps. The teaching indenters skip this. A quantitative model must not.

```

100     step = model.GeostaticStep(name=name, previous=previous,
101                                nlgeom=ON if nlgeom else OFF)

```

2 Implicit ALE

On top of implicit. examples/indenter/implicit_ale.

Reduced-integration first-order solids

ALE domains may contain only C3D8R (or CAX4R, CPE4R). Other types in the set are silently nonadaptive. The implicit model already uses C3D8R. Do not switch to fully integrated or second-order elements.

Adaptive mesh controls and a restricted set

A box written in absolute coordinates is the usual way to get an empty, inert domain. Print what the set actually contains.

```

51     ale.make_controls(model, 'ALE_BC', smoothing=UNIFORM)
52     box = indenter_model.ale_box(P)
53     n_adaptive, bbox = ale.box_element_set(asm, handles['soil_inst'],
54                                           'ALE_Box', box)
55     ale.describe_box_set('ALE_Box', n_adaptive, bbox, total=n_soil)
56     ale.add_domain(model, handles['step_name'], asm.sets['ALE_Box'], 'ALE_BC',
57                  frequency=P.ale_frequency, mesh_sweeps=P.ale_sweeps,
58                  initial_mesh_sweeps=None)

```

No initial mesh sweeps

Abaqus/Standard has none. Pass None so the keyword is omitted.

```

242     kwargs = dict(
243         region=region,
244         controls=controls,
245         frequency=frequency,
246         meshSweeps=mesh_sweeps,
247     )
248     if initial_mesh_sweeps is not None:
249         kwargs['initialMeshSweeps'] = initial_mesh_sweeps
250     return step.AdaptiveMeshDomain(**kwargs)

```

Diagnostics after *Static

CAE cannot write *Diagnostics. Inject it or you cannot prove the domain swept. Standard may still omit the Explicit-style .msg block.

```

63     inp = jobs.write_input(job, workdir)
64     inpedit.inject_diagnostics(inp, mode='summary', include_static=True)

```

Standard ALE is Lagrangian-only, one smoothing algorithm, intended for wear and acoustics. Do not expect skirt penetration from it.

3 Explicit

On top of implicit, replacing the solver. `examples/indenter/explicit`. This is the largest jump.

Mass (*Density)

Explicit cannot run without density. The stable increment is $L_{\min}/c_d$ and c_d depends on ρ . A Standard model that never had *Density will not data-check. This repository always writes it.

```

69     mat = model.Material(name=name)
70     mat.Density(table=((density,)),)
71     mat.Elastic(table=((youngs, poisson),))
72     return mat

```

Explicit element library

Same C3D8R code, EXPLICIT library. Mixing libraries in one part fails at data check.

```

150     library = EXPLICIT if solver == 'explicit' else STANDARD
151     # Hourglass control is left at the element-type default here. The
152     # explicit model can change it through *Section Controls.
153     elem = mesh.ElemType(elemCode=C3D8R, elemLibrary=library)

```

Dynamic explicit step, not static

Drop *Static and automatic stabilisation. Bulk viscosity is on by default on *Dynamic, Explicit (watch ALLVD). Do not add Rayleigh or material damping to hide kinetic energy. Slow the rate or scale less mass instead. See `docs/01-explicit-quasi-static.md`.

```

79     return model.ExplicitDynamicsStep(
80         name=name, previous=previous, description=description,
81         timePeriod=time_period,
82         nlgeom=ON if nlgeom else OFF,
83         improvedDtMethod=ON if improved_dt else OFF,
84     )

```

Time period from rate, not a unit interval

The step length is penetration depth divided by the prescribed velocity. That velocity is scaled far above reality. Print the increment count before you submit.

```

193     if solver == 'explicit':
194         time_period = P.ind_depth / float(P.ind_vel)
195         steps.explicit_step(model, step_name, 'Initial', time_period,
196                             nlgeom=True)
197         steps.smooth_ramp(model, 'Ramp', P.ramp_frac, time_period)

```

Smooth-step amplitude

A step change of velocity puts a stress wave through the model on increment 1. SMOOTH STEP ramps over the first few percent of the step.

```

152     from abaqusConstants import STEP
153     t1 = t_ramp * t_total
154     return model.SmoothStepAmplitude(name=name, timeSpan=STEP,
155                                     data=((0.0, 0.0), (t1, 1.0)))

```

Velocity control, not displacement

A constant rate makes the resistance curve interpretable. Sign is negative because penetration is $-z$.

```

222     model.VelocityBC(name='Push', createStepName=step_name,
223                      region=asm.sets['IndRP'],
224                      v1=0.0, v2=0.0, v3=-P.ind_vel,
225                      vr1=0.0, vr2=0.0, vr3=0.0,
226                      amplitude='Ramp', distributionType=UNIFORM)

```

Explicit general contact

ContactExp. Penalty friction. Mandatory later for CEL, and the right default here because the contact surface changes as the soil folds.

```

81     interaction = model.ContactExp(name='GeneralContact',
82                                   createStepName=step_name)
83     interaction.includedPairs.setValuesInStep(stepName=step_name,
84                                               useAllstar=True)
85     interaction.contactPropertyAssignments.appendInStep(
86         stepName=step_name, assignments=((GLOBAL, SELF, prop_name),))

```

Double precision job and domain parallel

Single precision corrupts a long soil run. Loop-level parallelisation is not available on Windows. numDomains must be a multiple of numCpus.

```

63     return mdb.Job(
64         name=job_name, model=model_name, description=description,
65         type=ANALYSIS,
66         explicitPrecision=DOUBLE_PLUS_PACK if double else SINGLE,
67         nodalOutputPrecision=FULL if double else SINGLE,
68         numCpus=n_cpus, numDomains=n_domains,
69         parallelizationMethodExplicit=DOMAIN,
70         multiprocessingMode=DEFAULT,

```

Energy history

Request it. Grade the peak ALLKE/ALLIE, not the final value. Below about 5 % is acceptable, below 1 % is quotable.

```

172     return model.HistoryOutputRequest(
173         name='Energies', createStepName=step_name,
174         variables=('ALLIE', 'ALLKE', 'ALLAE', 'ALLDC', 'ALLVD', 'ALLWK',
175                  'ALLPD', 'ALLSE', 'ETOTAL'),
176         numIntervals=interval,

```

Optional mass scaling

Semi-automatic, target Δt , throughout the step. It scales the worst elements, which in a penetration run are next to the structure. Check added mass in the `.sta` and re-check `ALLKE`.

```

128     when = THROUGHOUT_STEP if throughout else AT_BEGINNING
129     if target_dt is not None:
130         entry = (SEMI_AUTOMATIC, region, when, 0.0, target_dt, BELOW_MIN,
131                 1, 0, 0.0, 0.0, 0, None)
132     else:
133         entry = (SEMI_AUTOMATIC, region, when, factor, 0.0, None,
134                 1, 0, 0.0, 0.0, 0, None)
135     step.setValues(massScaling=(entry,))

```

Optional section controls

Distortion control or enhanced hourglass, not both. CAE in 2021 has no API. Inject after `*Preprint` and attach to the soil `*Solid Section`.

```

70     if case == 'C':
71         keyword = inpedit.section_controls_keyword(
72             'SC-Soil', distortion_control=True)
73         inpedit.inject_section_controls(inp, keyword, SOIL_SECTION)

```

UMAT becomes VUMAT

If the soil is a user material, the Explicit job needs a VUMAT (or a wrapper), the matching `*Depvar` count, and initialised state. Verify on `constitutive/single_element` first.

4 Explicit ALE

On top of explicit. `examples/indenter/explicit_ale`. Keep everything from the previous section.

Adaptive mesh controls

```

65     ale.make_controls(model, 'ALE_BC', smoothing=UNIFORM)

```

Restricted element set

Do not make the whole part adaptive. A box around the indenter is the production configuration. An empty set is an inert domain with no warning.

```

66     box = indenter_model.ale_box(P)
67     n_adaptive, bbox = ale.box_element_set(
68         asm, handles['soil_inst'], 'ALE_Box', box)
69     ale.describe_box_set('ALE_Box', n_adaptive, bbox, total=n_soil)
70     domain_elset = 'ALE_Box'
71     ale.add_domain(model, handles['step_name'], asm.sets['ALE_Box'],
72                  'ALE_BC', frequency=P.ale_frequency,
73                  mesh_sweeps=P.ale_sweeps,
74                  initial_mesh_sweeps=P.ale_initial_sweeps)

```

Initial mesh sweeps

Explicit has them. Default in CAE is 5 at the start of the step, then `frequency=10` and `mesh sweeps=1`. The keyword reference that says default sweeps is 5 is wrong. See `docs/02-ale-adaptive-meshing.md`.

Diagnostics, then prove it ran

Inject `*Diagnostics`, `adaptive mesh=summary` after `*Dynamic, Explicit`. Then read the `.msg`. Zero percent of nodes moved means the domain was defined and did nothing.

```
79 inp = jobs.write_input(job, workdir)
80 inpedit.inject_diagnostics(inp, mode='summary')
```

Parallel domains

Nodes shared between Explicit domains are nonadaptive. For a study of ALE, run one CPU. For production, several well-separated regions. Adjacent regions that share a face are treated as one. See `docs/03-ale-multi-region-parallel.md`.

Still not a remesh

Topology is fixed. When material must flow around a tip and close behind it, ALE stops helping. That is CEL.

5 Explicit CEL

On top of explicit, replacing the Lagrangian soil. `examples/indenter/explicit_cel`. Keep density, the Explicit step, smooth-step velocity drive, Explicit contact, double precision and energy output. Drop ALE.

Eulerian part with void above the soil

The mesh is a fixed box of space. Heave needs empty elements. No void means resistance is silently too high.

```
64 sk = model.ConstrainedSketch(name='eul_plan', sheetSize=10.0 * L)
65 sk.rectangle(point1=(-L / 2.0, -W / 2.0), point2=(L / 2.0, W / 2.0))
66 eul = model.Part(name='Eulerian', dimensionality=THREE_D, type=EULERIAN)
67 eul.BaseSolidExtrude(sketch=sk, depth=D + H)
68
69 if H > 0.0:
70     datum = eul.DatumPlaneByPrincipalPlane(principalPlane=XYPLANE,
71                                             offset=D)
72     eul.PartitionCellByDatumPlane(datumPlane=eul.datums[datum.id],
73                                  cells=eul.cells)
```

Eulerian elements

EC3D8R, Explicit library, structured hex. A swept or free mesh is rejected.

```
94 regions = (cells if cells is not None else part.cells,)
95 part.setElementType(regions=regions,
96                    elemTypes=(mesh.ElemType(elemCode=EC3D8R,
97                                             elemLibrary=EXPLICIT),))
98 part.setMeshControls(regions=regions[0], elemShape=HEX,
99                    technique=STRUCTURED)
```

Eulerian section, not a solid section

The soil section is `*Eulerian Section`. The indenter stays a Lagrangian solid.

```

55     key = instance_key or (material.lower().replace(' ', '_') + '-1')
56     section = model.EulerianSection(name=name, data={key: material})
57     return section, key

```

Initial volume fraction

Cells below $z = 0$ start full. Everything else starts empty. This is `*Initial Conditions`, `type=VOLUME FRACTION`.

```

126     filled = eul_inst.cells.getByBoundingBox(
127         zMin=-D - 1e-6, zMax=0.0 + 1e-6)
128     if len(filled) == 0:
129         raise RuntimeError('no Eulerian cells found below z = 0')
130     filled_set = asm.Set(name='SoilFilled', cells=filled)
131     cel.assign_material(model, asm, eul_inst, filled_set)
132     cel.check_void_fraction(asm, eul_inst, 'SoilFilled')

```

Walls, not rollers, and a free top

Zero normal displacement on the four sides and the base so material cannot leave the box. The top is open. Closing it is another way to lose heave.

```

169 def _eulerian_walls(model, asm, inst, L, W, D, H):
170     """Zero normal velocity on the four sides and the base of the Eulerian box."""
171     top = H
172     picks = {
173         'XMIN': ((-L / 2.0, 0.0, (top - D) / 2.0), 1),
174         'XMAX': ((L / 2.0, 0.0, (top - D) / 2.0), 1),
175         'YMIN': ((0.0, -W / 2.0, (top - D) / 2.0), 2),
176         'YMAX': ((0.0, W / 2.0, (top - D) / 2.0), 2),

```

No geostatic

`*Geostatic` has no Eulerian implementation. Initialise stress directly or settle dynamically (slow, and it rings).

EVF on the field output

Without Eulerian volume fraction you cannot see where the material is.

```

150     model.fieldOutputRequests['F-Output-1'].setValues(
151         variables=('S', 'PE', 'PEEQ', 'LE', 'U', 'V', 'A', 'RF', 'CSTRESS',
152                 'EVF', 'STATUS'),
153         numIntervals=20)

```

General contact is mandatory

The Eulerian–Lagrangian interaction exists only through general contact. There is no ALE block. The free surface is one element thick.

6 What to check in the deck

Rung	The deck must show
Implicit	<code>*Static, nlgeom=YES</code> , displacement BC, no <code>*Adaptive Mesh</code>
Implicit ALE	plus <code>*Adaptive Mesh</code> on a named elset, <code>*Diagnostics</code>
Explicit	<code>*Density, *Dynamic, Explicit, type=VELOCITY, ALLKE</code>
Explicit ALE	plus <code>*Adaptive Mesh, *Diagnostics</code> , nonzero nodes moved in the <code>.msg</code>
Explicit CEL	<code>*Eulerian Section, EC3D8R</code> , volume fraction, void, no ALE

Write the input. Read it. Assert. Then submit.